

Rotary Screw Air Compressor (AIR-COMP) Equipment Manual and SOP

STANDARD OPERATING PROCEDURE: ROTARY SCREW AIR COMPRESSOR Document ID: SOP-MNT-AIR-042 Asset Class: AIR-COMP Plant Location: Vindhya Precision Works, Plant Floor Building 3 Applicability: Maintenance Technicians, Reliability Engineers, Utility Operators Effective Date: January 15, 2026 Revision: 4.1

1. PURPOSE AND SCOPE

This standard operating procedure defines the operational baselines, preventive maintenance schedules, fault diagnostics, parts inventory, and safety-critical interventions for all rotary screw air compressors (Asset Class AIR-COMP) installed at Vindhya Precision Works. Compliance with these clauses is mandatory for all maintenance personnel.

2. NORMAL OPERATING RANGES AND ALARM SETPOINTS

All AIR-COMP units shall operate within the strict parameter thresholds defined in Table 2.1. Any excursion outside the warning threshold requires logged notification to the Maintenance Lead within 1 shift (8 hours). Excursion past the trip setpoint shall immediately shut down the compressor via the PLC safety interlock.

Table 2.1: Operating Parameters and Alarm Thresholds

Parameter	Normal Operating Range	Warning Setpoint	Trip Setpoint
Air Discharge Pressure	7.2 bar to 8.1 bar	8.4 bar	8.8 bar
Airend Discharge Temperature	78°C to 92°C	98°C	105°C
Sump Tank Pressure (Loaded)	7.0 bar to 7.9 bar	8.2 bar	8.7 bar
Main Drive Motor Current (415V)	110 A to 135 A	148 A	160 A
Vibration (Airend Drive End - Velocity)	0.8 mm/s to 2.2 mm/s RMS	3.5 mm/s RMS	4.8 mm/s RMS
Vibration (Airend Non-Drive End - Velocity)	0.6 mm/s to 1.8 mm/s RMS	3.0 mm/s RMS	4.2 mm/s RMS
Oil Injection Temperature	55°C to 68°C	72°C	78°C

Exception 2.1.1: During cold-start sequences in ambient temperatures below 12°C, oil injection temperature warning setpoints are temporarily bypassed for the first 300 seconds of operation.

3. PREVENTIVE MAINTENANCE SCHEDULE

Preventive maintenance (PM) is dictated by cumulative running hours recorded by the plant SCADA system. Technicians shall execute PM routines in accordance with Table 3.1.

Table 3.1: Running Hour PM Intervals

Interval	Task Identifier	Required Actions
Every 50 Hours	PM-50	Inspect condensate auto-drains on air receiver and moisture separator. Clean filter media on electrical cabinet cooling fans. Check oil level in sight glass.
Every 500 Hours	PM-500	Take compressor oil sample for laboratory spectrographic analysis. Grease main motor bearings with 25 grams of VPW-Polyrex EM grease. Inspect drive coupling alignment and elastomeric insert wear.
Every 2,000 Hours	PM-2000	Replace air intake filter element (Part No. VPW-AF-8821). Replace oil filter element (Part No. VPW-OF-4410). Inspect minimum pressure check valve seals. Clean oil cooler external fins with compressed air at max 2.0 bar.
Every 4,000 Hours	PM-4000	Replace air/oil separator element (Part No. VPW-OS-9930). Flush and refill compressor lubricant (VPW-Lube-Syn 46). Replace thermal valve element. Calibrate pressure transducers and temperature probes against certified secondary standards.
Every 16,000 Hours	PM-16000	Complete overhaul of airend assembly by certified technician. Replace shaft seals, main rotor bearings, and inlet control valve actuation diaphragms. Inspect gears/coupling for backlash tolerances.

4. FAULT-CODE TABLE AND TROUBLESHOOTING

When the compressor controller displays a fault code, technicians shall reference Table 4.1 to determine the probable cause and execute the prescribed corrective action.

Table 4.1: Fault Code Diagnostics

Fault Code	Display Text	Probable Cause	Corrective Action
F-101	*HIGH AIREND TEMP*	Low oil level in receiver tank; blocked oil cooler core; failing thermal valve wax element stuck closed.	Check oil sight glass and top up with VPW-Lube-Syn 46 if below 50%. Clean external oil cooler fins. Verify thermal valve opens at 65°C; replace element if defective.
F-102	*MOTOR OVERLOAD*	High discharge pressure setting; mechanical binding in airend bearings; low voltage supply phase imbalance exceeding 3%.	Inspect pressure switch settings. Rotate airend shaft manually (power off, locked out) to check for binding. Measure running current across all three phases; report supply imbalance to electrical utility team.
F-103	*LOW OIL PRESSURE*	Restricted oil filter; faulty oil pump bypass valve; collapsed internal oil line hose.	Replace oil filter element (VPW-OF-4410). Inspect oil stop valve and bypass valve for debris or stuck plungers. Replace any oil line showing internal delamination.
F-104	*HIGH DISCHARGE PRESS*	Pressure regulator failing; inlet unloader valve stuck open during unload cycle; PLC output relay latched.	Inspect and rebuild unloader valve assembly (Part No. VPW-UV-1102). Verify solenoid valve 3-way exhaust port is clear. Test PLC output channels.
F-105	*PHASE SEQ / LOSS*	Incorrect incoming phase rotation from motor control center; blown line fuse on control transformer.	Verify phase rotation using meter at main breaker (must be clockwise L1-L2-L3). Replace blown primary control fuses with 3A slow-blow type.
F-106	*SEPARATOR DP HIGH*	Air/oil separator element saturated with varnish or particulate contaminants; operating consistently below oil injection temperature dew point.	Replace air/oil separator element (VPW-OS-9930). Check that thermostatic valve maintains operating oil temperature above 65°C to prevent moisture condensation in sump.

5. CONSUMABLE AND SPARE PARTS LIST

Maintenance stockroom clerks shall maintain minimum stock quantities of the proprietary parts listed in Table 5.1. No substitutions are permitted without written sign-off by the Reliability Engineering Manager.

Table 5.1: Bill of Materials (BOM)

Description	Part Number	Stock Qty Threshold
Air Intake Filter Element	VPW-AF-8821	4 units
Spin-On Oil Filter Element	VPW-OF-4410	6 units
Air/Oil Separator Element	VPW-OS-9930	2 units
Synthetic Compressor Fluid (20L Pail)	VPW-Lube-Syn 46	3 pails
Thermal Valve Kit (65°C)	VPW-TV-6500	1 unit
Unloader Valve Repair Kit	VPW-UV-1102	1 unit
Minimum Pressure Valve Service Kit	VPW-MP-3320	1 unit
Airend Shaft Seal Kit	VPW-SS-5540	1 unit
Drive Coupling Elastomer Spider	VPW-DC-2210	2 units

6. TORQUE SPECIFICATIONS

All mechanical fasteners associated with the AIR-COMP asset class must be tightened using a calibrated torque wrench in accordance with Table 6.1. Thread-locking compound (Loctite 243 medium strength) shall be applied to steel-into-cast-iron fasteners unless otherwise noted.

Table 6.1: Fastener Torque Matrix

Application / Fastener Location	Thread Size	Target Torque
Airend Discharge Flange Bolts	M12 (Grade 8.8)	85 Nm
Main Drive Motor Mounting Bolts	M16 (Grade 8.8)	210 Nm
Oil Injected Manifold Pipe Flanges	M10 (Grade 8.8)	45 Nm
Minimum Pressure Check Valve Cover	M8 (Grade 8.8)	25 Nm
Air/Oil Separator Cover Fasteners	M10 (Grade 8.8)	50 Nm (Star pattern sequence)
Electrical Terminal Block Power Lugs	M8 Brass Studs	15 Nm

7. SAFETY-CRITICAL CORRECTIVE ACTIONS AND LOTO REQUIREMENTS

Certain maintenance interventions on Asset Class AIR-COMP involve extreme stored energy hazards, including high-pressure pneumatic air (up to 8.8 bar), elevated thermal energy (surfaces exceeding 100°C), and high-voltage electrical power (415V AC 3-phase).

7.1 Explicit List of Safety-Critical Actions The following corrective actions are strictly classified as Safety-Critical. Technicians are prohibited from initiating these tasks without executing an official Work Permit and applying Lockout/Tagout (LOTO) protocols per Vindhya Precision Works Safety Policy EHS-09:

1. Air/Oil Separator Element Replacement: Involves opening a pressure vessel containing residual pneumatic pressure and hot oil. Requires complete system blowdown, zero-pressure verification on gauge PI-02, and LOTO of the main electrical supply disconnect.
2. Airend Overhaul or Replacement: Involves uncoupling heavy rotating machinery, breaking flanged high-pressure lines, and handling heavy castings (>50 kg). Requires crane lift plan, mechanical LOTO, and electrical LOTO.
3. Thermal Valve or Minimum Pressure Valve Maintenance: Involves disassembly of housings connected directly to the oil receiver tank. Requires verified zero system pressure and thermal cool-down period below 40°C.
4. Main Drive Motor Maintenance or Alignment: Involves working near exposed rotating drive shafts and high-voltage terminals. Requires electrical LOTO at the Motor Control Center (MCC) bucket with personal padlock applied.
5. Oil Cooler Chemical Flushing or Core Replacement: Involves breaking hydraulic/oil circuits under residual head pressure. Requires closed-loop drainage, thermal isolation, and LOTO of the auxiliary scavenger pump circuit.

7.2 LOTO Verification Sequence Before any safety-critical task begins, the technician of record must: * Open the local isolator switch and apply a red padlock and danger tag. * Verify zero electrical voltage using a calibrated multimeter at the motor starter terminals. * Open the manual blowdown valve (V-99) to vent all receiver tank pressure to atmosphere. * Visually confirm that pressure gauge PI-01 reads exactly 0.0 bar before unbolting any piping flange or filter lid.